Vinay Kumar

Integrated PhD Research Scholar | Astrophysical Fluid Dynamics and Plasma Physics

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ICTS profile
the21vk.github.io
ORCID

Research interests

Magnetic reconnection; magnetohydrodynamic instabilities; fundamental plasma physics; astrophysical turbulence; direct numerical simulations; and high-performance scientific computing.

Education/Experience

2021–present

Integrated PhD in Theoretical Physics
International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India
Research area: astrophysical fluid dynamics and plasma astrophysics.

Summer 2026

Visiting PhD Fellow
Nordic Institute for Theoretical Physics (Nordita), Stockholm, Sweden

2018-2021

B.Sc. (Honors) Physics
Jamia Millia Islamia, New Delhi, India

Publications

(live list: the21vk.github.io/publications)

5. **Vinay Kumar** and Philipp Grete, "As a matter of reconstruction – varying the magnetic Prandtl number in implicit large eddy simulations," (2026). Submitted to arXiv, to be submitted to *The Astrophyical Journal*.

4. Axel Brandenburg and **Vinay Kumar**, "Evolutionary tracks and isochrones of the inverse Hall cascade," arXiv:2610.40092 (2026). Submitted, *Journal of Plasma Physics*.

3. **Vinay Kumar** and Axel Brandenburg, "Magnetic Prandtl number dependence of plasmoid-mediated reconnection," arXiv:2605.18946 (2026). Accepted, *Journal of Plasma Physics*.

2. **Vinay Kumar**, Rajaram Nityananda, and Joseph Samuel, "Closure invariants for polarised radio interferometric observations: a graph theoretical approach," *Journal of Astrophysics and Astronomy* 46(2) (2025).

1. **Vinay Kumar** and Pallavi Bhat, "Three-dimensional tearing instability of flux-tube-like magnetic fields," *Journal of Plasma Physics* 91, E99 (2025).

Awards and Fellowships

- Nordita Visiting PhD Student Fellowship (Summer 2026).
- TIFR Graduate School Fellowship (2021–2027).
- INSPIRE Scholarship for Higher Education (2018–2021).

Presentations

Conference presentations and posters

- Poster — *"As a matter of reconstruction – Steering the Magnetic Prandtl Number in Implicit LES,"* Plasma Turbulence Across Scales, ICTS, Bengaluru, 2026.

- Poster — *"Weak Magnetic Prandtl Number Dependence of Plasmoid Mediated Reconnection,"* Plasma Turbulence: New Challenges from Magnetic Confinement, ICTS, Bengaluru, 2026.
- Talk — *"Magnetic Reconnection in 3D and its Impact on Dynamos,"* International Symposium on Cosmic Magnetic Fields, NAOC, Beijing, 2025.
- Poster — *"Magnetic Reconnection in 3D and Its Role in Turbulent Dynamo Saturation,"* MIST: Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales, IESC, Corsica, 2025.
- Talk — *"Magnetic Reconnection and its Influence on Dynamos,"* Young Astronomers' Meeting (YAM), TIFR, Mumbai, 2025.
- Poster — *"Numerical and analytical investigations of 3D flux-tube magnetic reconnection,"* 42nd Meeting of the Astronomical Society of India, IISc, Bengaluru 2024.

Seminars

- *"Does MHD Turbulence Care About Its Prandtl Number? Plasmoids, Turbulent Decay, and Numerical Dissipation,"* Princeton Astroplasmas Seminar (online), Princeton University, New Jersey, 2026.
- *"Magnetic Reconnection in 3D: Tearing Instability of flux-tubes,"* Nordita Astrophysics Seminar, Nordita, Stockholm, 2026.
- *"Magnetic Reconnection in 3D: Tearing Instability of flux-tubes,"* Extragalactic Group Meeting, Hamburg Observatory, Hamburg, 2025.
- *"Magnetic Reconnection in 3D and its Impact on Dynamos,"* AIfA, Bonn, 2025.

Research visits

- Aalto University, Helsinki, 2026, Host: Prof. Maarit J. Korpi.
- University of Helsinki, Helsinki, 2026, Host: Prof. Joonas Nättilä.
- Nordita, Stockholm, 2026, Host: Prof. Axel Brandenburg.
  Supported by the Nordita Visiting PhD Student Fellowship.
- Hamburg Observatory, Hamburg, 2025, Host: Dr. Philipp Grete.
- Argelander Institute for Astronomy (AIfA), Bonn, 2025., Host: Prof. Jennifer Schober.

Academic service and outreach

- Teaching assistant for (i) Lab Course, (ii) Numerical Methods Course, and (iii) Research Methodology, Ethics and Communications Course at ICTS.
- ICTS In-house 2024: Organizer of an internal research-exchange meeting bringing together members of ICTS for talks, posters, and academic discussion.
- ICTS Open Day (2024,2025 and 2026): conducted outreach activities for the general public, including hands-on experiments and demonstrations of scientific concepts.
- Climate Chaos: We are Just Warming Up (2023): Co-developed interactive science exhibits on climate change for a public outreach exhibition at Science Gallery Bangalore, including demonstrations and visualizations communicating climate science to a general audience.

Technical skills

Large-scale codes used: Athena++, AthenaK, AthenaPK, Calliope, Dedalus, Parthenon, Pencil Code, PLUTO.
HPC systems used: JUPITER Booster (EuroHPC JSC), Dardel (PDC, KTH/Nordita), Rudra (SankhyaSutra Labs), Contra (ICTS-TIFR).
Languages: C/C++, Fortran, Python, Bash.